Experimental Theorems and Imported-Locator Ledger for RS-MCA

RS-MCA experimental notes

June 18, 2026

Abstract

This note is the experimental collection point for statements that are useful for the RS-MCA papers but are not yet main-paper theorems. It has two jobs. First, it records the correct provenance of the quotient-locator construction used in a separate AI-generated packet: the locator proof underlying item (d) is an imported theorem/proof pattern from Ben-Sasson–Carmon–Haböck– Kopparty–Saraf, *On Proximity Gaps for Reed–Solomon Codes* [1], especially their Theorem 7.1 and the Theorem 1.13 subgroup instantiation. That proof is not a new contribution of this repository. Second, it records genuinely local material that may be reusable later: the smooth-quotient notation dictionary, a list-fiber pigeonhole wrapper around the imported locator construction, the slack-two/subfield bookkeeping target for Paper D, and a new divisibility gate for Paper B's quadratic-extension, slack-two, three-co-support resonance window.

Everything here is experimental or imported unless explicitly promoted by a main paper. In particular, this note does not prove the corrected MCA conjecture, does not give a protocol consequence, and does not supersede Papers A–D.

1 Status, Provenance, and What Counts as New

This file is the place where we collect new things, but it must separate new repository contributions from imported constructions. The status labels used below are:

Imported	an external theorem, proof pattern, or conversion;
Wrapper	a repository repackaging, dictionary, or pigeonhole consequence;
New-local	a restricted algebraic theorem proved in this note;
Target	a precise proof obligation not yet promoted;
Heuristic	evidence or a search direction, not a theorem.

Hard attribution rule. Any locator-polynomial argument derived from the subgroup construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf must cite [1, Thm. 7.1] where the construction is stated or proved. Any admissible subgroup instantiation must also cite [1, Thm. 1.13]. The citation belongs at the theorem statement and again at the proof entry point; a bibliography entry or acknowledgement alone is not enough.

What is new in this note. The following items are the repository-side contributions collected here.

- A smooth-quotient notation dictionary translating BCHKS subgroup notation into the RS-MCA notation $Q = D^{c_{\mathrm{fib}}}$ and the dyadic/FFT-domain ledgers used in Papers A–D.

- A list-fiber pigeonhole wrapper: once a locator construction assigns many subset certificates to a smaller slope set, one heavy slope has many locator certificates, and, after a distinctness check, many nearby codewords.
- A slack-two and subfield-pigeonhole proof target for the Paper D route: the imported locator construction should be run at the augmented-code rung needed by the Crites–Stewart list-to-agreement conversion, and slopes should be pigeonholed over the generated/base field when the construction really takes values there.
- A New-local divisibility gate in the restricted Paper B window $B = \mathbb{F}_p$, $F = \mathbb{F}_{p^2}$, $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$. This is independent of the BCHKS priority issue and is the main theorem proved in Section 7.

What is not new. The polynomial identity behind the item-(d) locator proof is not new here. In BCHKS notation it is the expansion of a vanishing polynomial over a complement of a chosen subset; in smooth-quotient notation it is the same expansion of $\prod_{b\in A}(X^{c_{\mathrm{fib}}} - b)$, up to choosing a subset or its complement. The use of ℓ for the chosen subset size also follows the external presentation closely enough that it should either be cited or renamed to ℓ_{pg}.

Source map. The main papers remain the authoritative documents.

- Paper A, `tex/RS_disproof_v3.tex`, refutes the old no-slack smooth-domain support-wise line-MCA statement. If it uses the locator construction, it should cite BCHKS and call the repository contribution a smooth-domain/list-fiber corollary, not a new locator proof.
- Paper B, `tex/slackMCA_v3.tex`, is the main target for the local resonance material in Sections 4–10.
- Paper C, `tex/snarks_v4.tex`, records protocol ledgers. Nothing in this note may be consumed by a protocol without a separate conversion theorem.
- Paper D, `tex/cs25_cap_v4.tex`, gives the universal cap via an imported list-to-agreement route. Its locator-side input should cite BCHKS explicitly if it uses the quotient-locator construction.

2 BCHKS Locator Construction in Smooth-Quotient Notation

Status: Imported plus Wrapper. This section is a provenance-safe restatement and dictionary. It is included so later main-paper edits have a clean place to copy from.

2.1 Imported theorem/proof pattern

BCHKS prove a general multiplicative-subgroup construction [1, Thm. 7.1]. In a compact form, the imported statement is as follows.

Theorem 1 (BCHKS quotient-locator construction, imported)**.** *Let $G \subseteq H \subseteq \mathbb{F}_q^*$ be multiplicative subgroups and let $\Phi : H \to G$ be $x \mapsto x^c$, where $c = |H|/|G|$. Let $E \subseteq G$, and let $\ell_{\mathrm{pg}}, a \geq 1$ satisfy*

$$|E^{(+\ell_{\mathrm{pg}})}| = \#\{e_1 + \cdots + e_{\ell_{\mathrm{pg}}} : e_i \in E \ \textit{distinct}\} \geq a.$$

Let $D = \Phi^{-1}(E)$, $n = |D|$, *and*

$$C = \mathrm{RS}[\mathbb{F}_q, D, n - (\ell_{\mathrm{pg}} + 2)c].$$

Then, for $\gamma = \ell_{\mathrm{pg}} c/n$, *there are functions* $f, g : D \to \mathbb{F}_q$ *such that*

$$\#\{z \in \mathbb{F}_q : \Delta(f + zg, C) \le \gamma\} \ge a,$$

while

$$\Delta([f, g], C^2) \ge (\ell_{\mathrm{pg}} + 1)c/n.$$

Reference. This is exactly the imported construction of [1, Thm. 7.1]; the proof is not reproduced as a repository contribution. BCHKS define the line endpoints by the monomials $x^{n-\ell_{\mathrm{pg}} c}$ and $x^{n-(\ell_{\mathrm{pg}}+1)c}$, then expand a vanishing polynomial over the complement of a chosen ℓ_{pg}-subset. The coefficient of $X^{n-(\ell_{\mathrm{pg}}+1)c}$ is an affine transform of a restricted sum from $E^{(+\ell_{\mathrm{pg}})}$, giving the exceptional slope values. □

The BCHKS Theorem 1.13 is the admissible-subgroup specialization of this construction: take $E = G$, take $H = D$, set $c = |D|/|G|$, and use $\ell_{\mathrm{pg}} = |G|/2$. Any RS-MCA statement corresponding to that specialization should cite [1, Thm. 1.13] directly.

2.2 Notation dictionary

BCHKS notation	**RS-MCA notation**	**Meaning**
H	evaluation domain D	Large multiplicative domain on which codewords are evaluated.
G	quotient $Q = D^{c_{\mathrm{fib}}}$	Image of the power map; one point in Q has c_{fib} preimages in D.
$c = \vert H\vert/\vert G\vert$	$c_{\mathrm{fib}} = n/\vert Q\vert$	Fiber size of the quotient map. Avoid calling this a, because BCHKS also uses a for the number of exceptional slopes.
$\Phi(x) = x^c$	$x \mapsto x^{c_{\mathrm{fib}}}$	Quotient/fiber map.
$E \subseteq G$	active quotient subset $E \subseteq Q$	Quotient-level support used by the locator.
$E^{(+\ell_{\mathrm{pg}})}$	restricted first-coefficient/slope set	Sums of distinct chosen quotient elements; slopes are affine transforms of these sums.
ℓ_{pg}	ℓ_{pg} or renamed subset size	Number of chosen quotient elements. Use ℓ_{pg} in RS-MCA text to avoid collision with other ℓ's.
$f = x^{n-\ell_{\mathrm{pg}} c}$, $g = x^{n-(\ell_{\mathrm{pg}}+1)c}$	locator line endpoints	Same monomial line after quotient notation is substituted.
$P_S(X) = \prod_{\alpha \in S'}(X^c - \alpha)$	$\prod_{b \in A}(X^{c_{\mathrm{fib}}} - b)$, possibly after complementing A	The locator polynomial.
exceptional z's	bad slopes / heavy list fibers	Slope parameters for which a line word is close to the RS code.

2.3 The shared locator identity

The following identity is written down because it is the exact place where priority can be lost. The identity is the BCHKS identity in RS-MCA notation.

Lemma 1 (Locator expansion, imported identity)**.** *Let D map c_{fib}-to-one onto a quotient set E by $x \mapsto x^{c_{\mathrm{fib}}}$, and let $A \subseteq E$ have size ℓ_{pg}. Put $A^c = E \setminus A$ and define*

$$P_A(X) = \prod_{\alpha \in A^c} (X^{c_{\mathrm{fib}}} - \alpha).$$

Then

$$P_A(X) = X^{n-\ell_{\mathrm{pg}}c_{\mathrm{fib}}} - e_1(A^c)X^{n-(\ell_{\mathrm{pg}}+1)c_{\mathrm{fib}}} + \textit{terms of degree at most } n - (\ell_{\mathrm{pg}} + 2)c_{\mathrm{fib}}.$$

Equivalently, since $e_1(A^c) = e_1(E) - e_1(A)$, the coefficient of the next monomial is an affine transform of the restricted sum $e_1(A)$.

Reference. This is the elementary expansion used inside [1, Thm. 7.1]. We record it only to fix notation. It should not be cited as a new lemma of the repository unless the BCHKS provenance is present at the statement. □

2.4 List-fiber pigeonhole wrapper

The next proposition is the repository-side wrapper around the imported identity. It is intentionally stated in certificate form, because distinctness of the nearby codewords is a separate check in concrete instantiations.

Proposition 1 (List-fiber pigeonhole wrapper)**.** *Let $\mathcal{A}$ be a family of ℓ_{pg}-subsets of a quotient set E. Suppose the BCHKS locator construction assigns to each $A \in \mathcal{A}$ a slope $z(A)$ lying in a finite set S, and a locator certificate showing that the corresponding word $f + z(A)g$ is within radius γ of a locator codeword $P_A \in C$. Then some slope $z_0 \in S$ has at least*

$$\frac{|\mathcal{A}|}{|S|}$$

locator certificates above it. If the codewords P_A attached to the certificates over z_0 are distinct, then the Hamming ball of radius γ around $f + z_0 g$ contains at least $|\mathcal{A}|/|S|$ codewords of C.

Proof. The first assertion is the pigeonhole principle applied to the map $A \mapsto z(A)$. The second assertion is only the definition of list size, after the distinctness check removes possible duplicate locator codewords. The construction that supplies the certificates is imported from Theorem 1. □

Remark 1 (What the wrapper adds)**.** *The new part here is not the locator polynomial. The wrapper isolates the extra bookkeeping needed by RS-MCA: which slope set pays the pigeonhole bill, whether the slope set is the ambient field $\mathbb{F}$ or a generated/base subfield B, and whether locator certificates remain distinct after specialization. Those are the checks needed before using the construction in Papers A, B, or D.*

2.5 Slack-two and subfield target for Paper D

Status: Target. The Paper D route needs a locator-heavy list fiber in the augmented code rung $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$, because that is the object fed into the Crites–Stewart list-to-agreement conversion. The proof obligation should be stated as follows.

Proposition 2 (Slack-two/subfield locator target)**.** *Let $D \subseteq B \subseteq \mathbb{F}$, and suppose a BCHKS-type quotient locator construction over D produces slopes in B, not merely in $\mathbb{F}$. If the one-rung/augmented-code version produces distinct locator codewords in $C^+ = \mathrm{RS}[\mathbb{F}, D, k+1]$ for a family $\mathcal{A}$ of certificates, then there exists a slope $z_0 \in B$ with list size at least*

$$\frac{|\mathcal{A}|}{|B|}$$

inside the appropriate Hamming ball for C^+.

Bookkeeping only. This is Proposition 1 with $S = B$. The nontrivial work is external or elsewhere: one must verify the BCHKS locator hypotheses at the augmented-code rung, verify that the slope coefficients lie in B, and verify distinctness of the produced codewords. Until those checks are present, this proposition is a target, not a promoted Paper D theorem. □

2.6 Pasteable main-paper wording

Paper A wording. The polynomial locator identity used here is the quotient-fiber form of the construction of Ben-Sasson–Carmon–Haböck–Kopparty–Saraf [1, Thm. 7.1]; their Theorem 1.13 is the admissible subgroup instantiation. Thus the locator proof itself is not a new contribution of this manuscript. The present use is the smooth-domain specialization, the field/divisor bookkeeping, and the list-fiber pigeonhole packaging.

Paper D wording. The locator-side input is the BCHKS quotient-locator construction [1, Thm. 7.1; cf. Thm. 1.13], written in the smooth-domain quotient notation $Q = D^{c_{\mathrm{fib}}}$. The additional bookkeeping needed here is the slack-two augmented-code rung and, when available, subfield pigeonholing over B rather than over the ambient extension field $\mathbb{F}$.

3 Triage of the AI-Generated Four-Item Packet

Status: Audit. The four advertised items should be positioned as follows before any promotion into a main paper.

Item	Limitation	Repository handling
(a)	Works only at slack on the order of $1/\sqrt{n}$.	Use as a weak-slack comparison point only. It does not clear the corrected-reserve MCA objective.
(b)	Applies only to three Fermat-prime cases.	Treat as finite-prime evidence or a worked computation, not as a uniform smooth-domain theorem.
(c)	Requires exponential field size.	Treat as a large-field sanity check. It is not directly usable for polynomial-field or prize-ledger parameters.
(d)	Most relevant, but the locator proof is essentially BCHKS.	Import, cite, and adapt from [1, Thm. 7.1 and Thm. 1.13]. The repository contribution is only the smooth-domain/list-fiber/slack-two packaging after exact hypotheses are checked.

Promotion checklist. Before any statement using item (d) moves to Papers A–D, check the following.

- Does the statement cite BCHKS at the theorem and proof entry point?
- Is the object a proximity-gap statement, a list statement, CA, MCA, line-decoding, or a protocol-facing theorem?
- Which field pays the slope/list pigeonhole bill: B, $\mathbb{F}$, or another generated field?
- Are the locator codewords distinct after specialization?
- Is the claimed slack in the same normalization as the consumed theorem?
- Are items (a), (b), and (c) kept out of asymptotic/protocol claims unless their limitations are explicitly part of the theorem statement?

4 Restricted Resonance Setup for Paper B

Status: New-local for the divisibility gate; otherwise local normal-form bookkeeping. The remaining sections preserve and reorganize the Cycle 14–18 algebraic material. This branch is independent of the BCHKS priority correction.

The common restricted setting is

$$B = \mathbb{F}_p, \qquad F = \mathbb{F}_{p^2}, \qquad D = \mathbb{F}_p, \qquad t = \sigma = 2, \qquad j = 3,$$

and we work off the tangent/global endpoint

$$R_0 = \{\, [W]_E \wedge [B_{\mathrm{num}}]_E = 0 \,\}.$$

The field ledgers remain distinct:

$$q_{\mathrm{gen}} = p, \qquad q_{\mathrm{line}} = p^2.$$

The regime is sub-reserve, since $\eta = \sigma/n = 2/n$. Thus none of the following proves a corrected-reserve list, CA, MCA, line-decoding, SNARK, or protocol theorem.

The base/generated field is $B = \mathbb{F}_p$. The line field is the quadratic extension $F = \mathbb{F}_{p^2} = B(\alpha)$, with basis $\{1, \alpha\}$ over B and $\alpha^2 = \nu \in B$. The domain is $D = \mathbb{F}_p$, so $n = |D| = p$.

A degree-two residue-line datum consists of

$$E \in F[X], \qquad \deg E = 2, \qquad E(x) \neq 0 \text{ for all } x \in D,$$

and a numerator

$$B_{\text{num}} \in F[X], \qquad \deg B_{\text{num}} < 2.$$

We write

$$A = F[X]/E, \qquad [P]_E \in A, \qquad b = [B_{\text{num}}]_E.$$

The anchor word on D is interpolated by a polynomial W. The operation $u \wedge v$ means the determinant of the coordinate vectors of $u, v \in A$ in a chosen F-basis of the two-dimensional F-space A. Off R_0, the pair $\{[W]_E, b\}$ is such a basis.

The co-support has size $j = 3$. If $T \subset D$ is the three-point co-support, define

$$\tau_1 = e_1(T), \qquad \tau_2 = e_2(T), \qquad \tau_3 = e_3(T).$$

All valid co-supports have $\tau_i \in B$. The quantity C_2 denotes the number of distinct slopes produced by the relevant landing branch after restricting to valid split triples $T \subset \mathbb{F}_p$. Bounding landing points therefore bounds C_2, but a sharp slope theorem may require more because many landing points can share a slope.

5 Cycle 14 Normal Form

The Cycle 14 audit reduces the $t = 2, j = 3$ landing equation to two affine forms in the elementary co-support parameters

$$\tau = (\tau_1, \tau_2, \tau_3).$$

Lemma 2 (Affine wedge normal form)**.** *Assume $\{[W]_E, b\}$ is an F-basis of A. Then the landing equation can be written in the form*

$$\iota(\tau) = A_0(\tau_1, \tau_2) - \tau_3[W]_E, \qquad \mu(\tau) = B_0(\tau_1, \tau_2) - \tau_3 b,$$

where

$$A_0 = p_1[W]_E + p_2 b, \qquad B_0 = q_1[W]_E + q_2 b,$$

and $p_i, q_i \in F$ are affine-linear functions of (τ_1, τ_2). If the basis is normalized so $[W]_E \wedge b = 1$, then

$$\Delta = \iota \wedge \mu = (p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

Proof. The Cycle 14 quotient calculation, which is a concrete instance of Paper B's residue-line normal form, gives the displayed affine forms. Expanding in the ordered basis $\{[W]_E, b\}$, the coordinate vector of ι is $(p_1 - \tau_3, p_2)$, while the coordinate vector of μ is $(q_1, q_2 - \tau_3)$. Their wedge determinant is therefore

$$(p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

□

6 Cycle 18 Component Split

Choose a basis $F = B + \alpha B$ with $\alpha^2 = \nu \in B$, and write

$$p_1 = p_{10} + \alpha p_{11}, \quad p_2 = p_{20} + \alpha p_{21}, \quad q_1 = q_{10} + \alpha q_{11}, \quad q_2 = q_{20} + \alpha q_{21}.$$

Split

$$\Delta = \Delta_0 + \alpha \Delta_1, \qquad \Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3].$$

Theorem 2 (Monic quadratic and linear alpha component)**.** *In the setting above,*

$$\begin{aligned}\Delta_0 &= \tau_3^2 - (p_{10} + q_{20})\tau_3 + p_{10}q_{20} + \nu p_{11}q_{21} - p_{20}q_{10} - \nu p_{21}q_{11},\\ \Delta_1 &= -(p_{11} + q_{21})\tau_3 + p_{10}q_{21} + p_{11}q_{20} - p_{20}q_{11} - p_{21}q_{10}.\end{aligned}$$

Consequently Δ_0 is monic quadratic in τ_3, while Δ_1 has degree at most one in τ_3.

Proof. Substitute the coordinate expressions for p_i, q_i into the identity of Lemma 2 and use $\alpha^2 = \nu$. Collecting the coefficient of 1 gives the displayed formula for Δ_0, and collecting the coefficient of α gives the displayed formula for Δ_1. The coefficient of τ_3^2 in Δ_0 is 1, and the α-component has no τ_3^2-term. □

Corollary 1 (Graph reduction outside the zero-alpha branch)**.** *If Δ_1 is not the zero polynomial, write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2).$$

On the open locus $s \neq 0$, the common-zero condition $\Delta_0 = \Delta_1 = 0$ is supported on the graph

$$\tau_3 = -h/s.$$

Proof. This is immediate from solving the linear equation $\Delta_1 = 0$ for τ_3 on $s \neq 0$. □

7 New Divisibility-Gate Theorem

The next theorem is the main New-local theorem extracted from the Cycle 18 reduction. It is stronger than merely saying that the branch is a graph: it shows that any two-dimensional family must satisfy an explicit divisibility identity.

Theorem 3 (Divisibility gate for the graph branch)**.** *Assume the restricted setting above and assume $\Delta_1 \neq 0$. Write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2),$$

and define the denominator-cleared graph polynomial

$$G(\tau_1, \tau_2) = s(\tau_1, \tau_2)^2 \Delta_0(\tau_1, \tau_2, -h(\tau_1, \tau_2)/s(\tau_1, \tau_2)).$$

Then $\deg G \leq 4$. If G is not the zero polynomial, then

$$\#\{\, \tau \in B^3 : \Delta_0(\tau) = \Delta_1(\tau) = 0 \,\} = O(p).$$

Consequently the corresponding number of distinct slopes satisfies

$$C_2 = O(p) = O(n).$$

Therefore a growing-prime family with

$$C_2 = \Theta(p^2) = \Theta(q_{\text{line}})$$

in the nonzero-Δ_1 branch can occur only if $G \equiv 0$.

Proof. Because the p_i, q_i are affine-linear in (τ_1, τ_2), the coefficient s has degree at most 1, and h has degree at most 2. Since Δ_0 is monic quadratic in τ_3, clearing the denominators after substituting $\tau_3 = -h/s$ gives a polynomial G of degree at most 4.

On $s \neq 0$, the equation $\Delta_1 = 0$ forces $\tau_3 = -h/s$. Hence every common zero on this open locus gives a base pair (τ_1, τ_2) with $G(\tau_1, \tau_2) = 0$. If G is nonzero, Schwartz–Zippel gives at most $4p$ such base pairs, and each base pair gives at most one value of τ_3.

It remains to count the exceptional locus $s = 0$. There $\Delta_1 = 0$ also forces $h = 0$. If s is a nonzero affine form, this is contained in an affine line; if $s \equiv 0$, then h is a nonzero polynomial of degree at most 2. In either case there are $O(p)$ base pairs. For each such pair, Δ_0 is a monic quadratic in τ_3, so it gives at most two values of τ_3. The total common-zero count is $O(p)$.

Finally, the number of distinct slopes is bounded by the number of landing points τ. Thus $C_2 = O(p)$ whenever $G \not\equiv 0$, and any $\Theta(p^2)$ family in this branch must satisfy $G \equiv 0$. □

Remark 2. *The identity $G \equiv 0$ says that, after clearing the s-denominator, Δ_0 is divisible by the linear equation Δ_1 on the graph branch. This is the first exact algebraic gate that a large slope family in this restricted window must pass.*

8 Nearby Proven Gates

The same restricted lane contains two other useful gates. They are included here because they explain where Theorem 3 fits.

Proposition 3 (Base-component complete-intersection gate). *Suppose $\Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3]$ have no common surface component. Then*

$$\#V(\Delta_0, \Delta_1)(B) = O(p),$$

and hence this branch contributes $C_2 = O(p)$ slopes.

Proof. The two components have bounded degree. If they share no surface component, their common zero set in $\mathbb{A}^3_B$ is a curve of bounded degree. A bounded-degree affine curve over $\mathbb{F}_p$ has $O(p)$ rational points, and slopes are bounded by landing points. □

Proposition 4 (Cycle 16 determinant gate). *Let $Q(z_0, z_1)$ be the Cycle 15–16 determinant consistency polynomial for the affine system*

$$L_z(\tau) = \iota(\tau) - z\mu(\tau) = 0, \qquad z = z_0 + \alpha z_1.$$

If Q is not identically zero, then in the restricted $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$, off-R_0 window,

$$C_2 \leq 4p = O(p) = O(n).$$

Proof. Cycle 16 gives $\deg Q \leq 4$ and shows that every landing slope must satisfy $Q(z_0, z_1) = 0$. If $Q \neq 0$, Schwartz–Zippel over $B = \mathbb{F}_p$ gives at most $4p$ possible pairs (z_0, z_1), hence at most $4p$ slopes. □

9 Slope Map and Remaining Heuristics

On the graph branch, the landing equation $\iota = z\mu$ gives

$$p_1 - \tau_3 = zq_1, \qquad p_2 = z(q_2 - \tau_3).$$

Where $q_1 \neq 0$, substituting $\tau_3 = -h/s$ yields

$$z(\tau_1, \tau_2) = \frac{p_1 + h/s}{q_1}.$$

Equivalently, eliminating τ_3 recovers the Cycle 14 quadratic

$$q_1 z^2 - (p_1 - q_2)z - p_2 = 0,$$

so the residual map is controlled by

$$(\tau_1, \tau_2) \longmapsto [\, q_1 : (p_1 - q_2) : p_2 \,] \in \mathbb{P}^2(F).$$

Heuristic 1 (Exceptional identity should be rare)**.** *For generic source data, the polynomial G should not vanish identically. Thus the graph branch should usually be curve-sized, and the first possible large-slope families should come from algebraically special data satisfying the divisibility identity $G \equiv 0$ or from the zero-alpha branch $\Delta_1 \equiv 0$.*

Heuristic 2 (What remains after the gate)**.** *If $G \equiv 0$, the problem is no longer to show that the common-zero set is small. It may be surface-sized. The remaining question is whether the projective coefficient map has one-dimensional image on every source-valid surface, giving $C_2 = O(p)$, or whether some growing-prime source-valid family has two-dimensional image and $C_2 = \Theta(p^2)$.*

10 Next Experimental Checks

A useful scanner should compute, for each source-valid candidate,

$$\Delta_1 = s\tau_3 + h, \qquad G = s^2 \Delta_0(\tau_1, \tau_2, -h/s),$$

then record whether $G \equiv 0$, whether $\Delta_1 \equiv 0$, the exceptional locus $s = h = 0$, the image size of

$$[\, q_1 : (p_1 - q_2) : p_2 \,],$$

and the direct slope count over distinct split triples $T \subset \mathbb{F}_p$.

The certificate should keep at least

$$p,\ q_{\text{gen}},\ q_{\text{line}},\ E,\ B_{\text{num}},\ \text{seed},\ \text{stratum},\ R_0\text{-status},$$
$$\Delta_1 \equiv 0, \quad G \equiv 0, \quad \#\{s = h = 0\}, \quad \#\,\text{image}, \quad C_2, \quad \#\{T \text{ split distinct}\}.$$

11 Non-Claims

This note does not claim:

- a proof of the corrected MCA conjecture;
- a positive theorem above corrected reserve;

- a q_{gen} local-limit theorem;
- a list, CA, MCA, line-decoding, SNARK, or protocol consequence;
- a large $\Theta(p^2)$ counterpacket;
- priority for the BCHKS quotient-locator proof pattern.

The mathematical value is narrower. On the imported side, the file gives a provenance-safe dictionary and wrapper for using BCHKS without claiming it as new. On the local Paper B side, the possible large branch in the restricted quadratic-extension window must satisfy an explicit algebraic identity before it can even be surface-sized.

A BibTeX Entry to Copy into Main Papers

```
@misc{BCHKS25,
  author = {Eli Ben-Sasson and Dan Carmon and Ulrich Habock and
            Swastik Kopparty and Shubhangi Saraf},
  title  = {On Proximity Gaps for Reed--Solomon Codes},
  year   = {2025},
  note   = {Manuscript dated November 11, 2025},
  url    = {https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf}
}
```

References

[1] Eli Ben-Sasson, Dan Carmon, Ulrich Haböck, Swastik Kopparty, and Shubhangi Saraf. *On Proximity Gaps for Reed–Solomon Codes*. Manuscript dated November 11, 2025. https://www.math.toronto.edu/swastik/rs-proximity-gaps-2025.pdf.